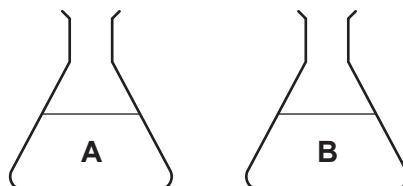


8. Hard water can be softened using different methods, including boiling the water and passing it through an ion exchange column.

- (a) Two hard water samples, **A** and **B**, were tested for hardness, using soap solution. Both samples were tested before and after boiling, and after passing through an ion exchange column.



	Volume of soap solution needed to produce a lather (cm ³)		
Water sample	Before boiling	After boiling	After ion exchange
A	15	15	2
B	17	9	2

State the type (or types) of hard water that each sample contains. Give your reasoning. [4]

A

.....

.....

B

.....

.....

.....

.....



- (b) Sodium carbonate (washing soda) can also be used to soften hard water. It reacts with the magnesium compounds in the water.

Write a balanced symbol equation for the reaction between sodium carbonate and magnesium chloride, to form sodium chloride and magnesium carbonate. [3]

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Examiner
only

7

